

PHY180 Final Report - Analysis of a Simple Pendulum

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1. INTRODUCTION

The purpose of this experiment is to analyze the relationship between period, damping (Q-factor), and adjustable length using a handmade simple pendulum to compare with theoretical models. The simple pendulum theoretically acts like simple harmonic motion in physics, where motion at small-angles can be assumed to be proportional to momentum and displacement. However, in real experiments, various obstacles such as air resistance, friction, and damping by string tension, which occurred, damped the oscillation. This experiment analyzes the damped harmonic oscillation and period changes of a simple pendulum by comparing this theoretical model and the actual pendulum's motion.

The motion of a simple pendulum is a simple harmonic motion recovered by gravity, and the general time-dependent angular displacement of a pendulum can be expressed by the following equation.

$$\theta(t) = \theta_0 e^{-t/\tau} \cos\left(2\pi \frac{t}{T} + \phi_0\right) \quad [1]$$

Here, θ_0 is the initial angle, τ is the decay constant, T is the period, and ϕ_0 is the initial phase. The equation describes a damped harmonic oscillation in which the amplitude decreases exponentially with time. It is used to fit the Time vs. Amplitude graph.

For these experiments, τ is calculated of 178 ± 1 s.

To quantitatively calculate this damping, the Q-factor is defined using the following equation.

$$Q = \pi \frac{\tau}{T} \quad [2]$$

The Q-factor indicates how efficiently a pendulum conserves energy; a higher Q-factor results in less damping with long oscillations.

In this experiment, τ was determined through Time vs. Amplitude graph fitting, and the Q-factor was calculated with the measured average period T .

In addition, the following expression was used to measure the Q-factor by a manual approach.

$$Q = 4N \quad [3]$$

Here, N is the number of cycles until the amplitude decreases to approximately 46% ($e^{-\pi/4}$) of its initial value. This method calculated Q using the number of cycles measured by manually counting each cycle. The Q-factor from both methods matches within the uncertainty, which indicates the accuracy of the damping model.

The period of a pendulum can vary with its initial angle, and this is expressed by a polynomial approximation.

$$T = T_0(1 + B\theta_0 + C\theta_0^2 + \dots) \quad [4]$$

Here, T_0 is the period in the small-angle approximation, B represents the asymmetry of the system, and C represents the rate of increase in the period by the square of the angle. For simple harmonic motion, B should be 0, and if $C > 0$, it means the period increases as the release angle also increases. In experiment 1, B and C were derived by using this expression and confirming that the small-angle approximation applies only for angles below ± 0.17 rad ($= 9.74^\circ$).

The relationship between the length and period of a pendulum is expressed by a Power Law Model.

$$T = kL^n \quad [5]$$

where k is the constant with units of $\text{s} \cdot \text{m}^{-n}$, and n is the power exponent; n is predicted to be 0.5 and k is to be 2.0. This equation is just derived from the basic simple harmonic motion equation below.

$$T \approx 2\pi \text{ s m}^{-1/2} \sqrt{L} \quad [6]$$

Quantitatively, the experimental results show a power law exponent of $n = 0.43 \pm 0.05$ and a constant $k = 1.94 \pm 0.02$ for the period length relation. Which n has a way lower value than the theoretical prediction of $n = 0.5$ with 13.6% differences. In contrast, the measured value of k differs from the theoretical value by only 2.9%, remaining relatively close within the experimental uncertainty.

Finally, the relationship between the Q-factor vs. Length graph was approximated by the following linear function.

$$Q(L) = aL + b \quad [7]$$

a represents the rate at which the Q-factor increases with length, and b represents the theoretical damping constant when the length is zero. This function explains that for short pendulums, resulting in a low Q-factor, and as the length increases, the Q-factor gradually increases.

In this experiment, the theoretical model and experimental results were compared to evaluate the accuracy, and the main factors that affect the real pendulum motion were also examined. The results showed that the small-angle approximation remained constant, and the relationship between period and length also followed the theoretical model. Overall, the experimental outcomes were consistent with theoretical expectations, confirming that the adopted model provides a reliable description of the simple pendulum's motion.

2. METHODS AND PROCEDURES

Before starting the actual measurements, the measurement uncertainties associated with the experimental apparatus were verified. A rigid ruler with a minimum scale division of 1 mm was used for all length measurements to prevent bending or stretching that could introduce additional errors. Therefore, a Type B uncertainty of $\pm 0.5 \text{ mm}$ ($= 0.05 \text{ cm} = 0.0005 \text{ m}$) was applied to all measured lengths.

Similarly, to ensure consistent angle measurements without relying on digital tools that could introduce time delays or calibration errors, a plastic protractor with a minimum scale division of 1° was used. The corresponding Type B angular uncertainty of $\pm 0.5^\circ$ ($= 0.009 \text{ rad}$) was applied to all angle measurements.

This uncertainty and the reasoning for choosing the measurement tools were consistently reflected throughout the experiment.

2.1. Experimental Setup

To proceed with this experiment, we constructed a simple pendulum apparatus designed to minimize mechanical friction and ensure structural stability. This apparatus was made using a translucent acrylic wrist rest, a plastic protractor, regular thread, double-sided tape, and a metal keychain used as the weight (bob) (Figure 1).

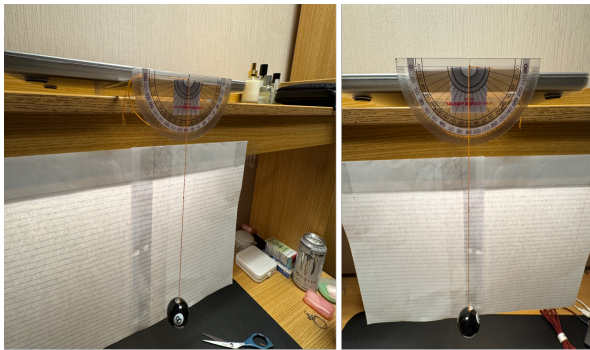


FIG. 1. Variety of views of the pendulum that was constructed for this lab.

The palm rest forms the base structure of the pendulum, measuring $28.4 \text{ cm} \times 9.7 \text{ cm} \times 1.5 \text{ cm}$. The protractor was attached to the center of the palm rest's side using double-sided tape. A crucial point is that a notch was carved at the center point with a utility knife before attaching the protractor. This design choice was essential for creating a clear rotational axis, allowing the string's axis of rotation to perfectly align with the protractor's origin point. This alignment minimized parallax error during angle measurements.

This string was designed to be adjustable in length at six different points (spaced at intervals of 0.05 m) and secured with knots. Using knots instead of sliding clips eliminated uncertainty from slippage or inconsistent length

measurements between experiments, enabling clear and repeatable length settings. These knots were inserted into the protractor's slots for quick adjustment.

The weight used is a spherical metal weight with a mass of approximately 70 g . A high-density metal material was selected to maximize the mass-to-surface-area ratio, making air resistance negligible compared to gravity. Furthermore, the spherical shape ensures a clearly defined center of mass, which is crucial for accurate length measurement. The thread was passed through the top hole of the bob, and the final length was measured to be $22.10 \pm 0.05 \text{ cm}$. This value represents the distance from the pivot to the center of mass ($20.00 \pm 0.05 \text{ cm}$ for the string and $2.10 \pm 0.05 \text{ cm}$ for the radius of the bob), and strictly follows the theoretical definition of a simple pendulum.

Finally, to control environmental variables and to ensure a rigid reference frame, a laptop weighing approximately 1.5 kg was placed on top of the pendulum base. This additional mass held the base firmly in place, preventing the phenomenon of energy transfer from the pendulum to the base (resonance) and eliminating measurement errors caused by base instability during vibrations.

2.2. Experimental Procedure

To obtain the Period vs. Angle graph, we measured the period of the pendulum at a total of 16 different angles: from ± 0.17 to 1.4 radians ($\pm 10^\circ$ to 80°) with intervals of 0.17 radians (10°). We repeated the experiments five times at each angle and computed the average value to reduce uncertainty. For data collection, an iPhone 15 Pro was used to record in HD quality at 60 fps for each angle, and one complete cycle was recorded. Subsequently, the IINA video player software was used to measure each oscillation period precisely, with its capability to display frames down to 1 millisecond intervals. Furthermore, all measured angles were converted to radians for analysis.

After generating the Period vs. Angle graph, we proceeded to obtain the Time vs. Amplitude graph. Initially, we set a specific angle to release the pendulum and tracked its movement until it stopped to determine the Q-factor. For this, the initial release angle was set to 30° (0.52 rad) (FIG. 2), and the entire process was recorded until the pendulum's motion gradually decayed to a stop. Based on the recorded video, the pendulum's motion was analyzed and tracked frame-by-frame using Tracker, an automatic tracking software.

The simple pendulum continued its motion for approximately 13 minutes from beginning to end. However, tracking the entire motion frame-by-frame using the Tracker software proved difficult due to the large number of frames and the resulting data volume. To address this, the video frame was down to 30 fps , and the analysis was performed only on the first 200 -second interval of the recording. Although the full interval could not be tracked, the reduced dataset was sufficient to obtain the graphs used in this experiment.

Time vs. Amplitude graph exhibited the form of an ideal ex-

ponential decay curve, consistent with the damped harmonic oscillator model described in Eq. [1].

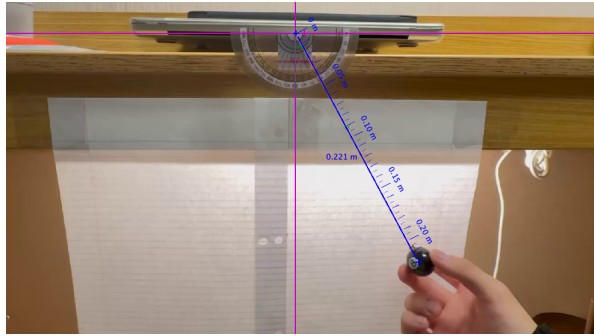


FIG. 2. Experimental setup of the pendulum showing the initial release angle of 0.520 ± 0.009 rad with a string length of 0.2210 ± 0.0005 m.

In the second part of the experiment, we have investigated the dependence of the oscillation period on the length of the pendulum.

The adjustable length was changed in intervals of 0.05 m, giving a total of six data points from 0.05 m to 0.30 m. To avoid the effect of varying mass or string properties, the material was kept constant for all configurations. For each length, the pendulum was released from an initial angle of 0.52 rad ($= 30^\circ$), and five complete oscillations were recorded. The oscillation periods were measured using the IINA video player software, and the average period was calculated. The recorded data were then used to generate the Period vs. Length graph, and the Power Law Model in Eq. [5] was applied to fit the graph.

In the Power Law Model, Eq. [5], T represents the oscillation period, L the pendulum length, k the proportionality constant, and n the power exponent in the power law relationship.

After determining the relationship between period and length, a log-log plot was generated using the same data to examine the power law behavior.

The data were plotted on log scales for both axes, and a linear regression was applied to obtain the slope and intercept.

The slope of the best-fit line represents the experimental value of the exponent n , and the intercept corresponds to the proportional constant k in the Power Law Model, Eq. [5].

Lastly, to determine the Q-factors for every six different lengths, the same method that was used in the Time vs. Amplitude section was repeated six times for each length to investigate how damping behavior varies.

For each length configuration, the decay constant τ was extracted using a Python-based nonlinear curve-fitting (generated with the assistance of AI tools^[8]), in which the data were fitted to the damped harmonic oscillator model given in Eq. [1].

This process provided a set of Q values corresponding to each pendulum length, and then compiled into a single dataset to make the Q-factor vs. Length graph.

3. RESULTS AND ANALYSIS

Based on the collected data, the standard deviation was calculated for each angle. This was used to determine the uncertainty and applied to the graph. Additionally, for Type B uncertainty, $\pm 0.5^\circ$ was applied, which is half the minimum scale division (1°) of the protractor. This was converted to its radian value of $0.5^\circ \times \frac{\pi}{180^\circ} \approx 0.009$ rad for use.

3.1. Period vs. Angle

The Period vs. Angle graph (FIG. 3) shows a graph of the pendulum's period as a function of the initial release angle. The data are presented with uncertainty, and a quadratic curve of best fit is shown. This indicates that the period increases as the angle increases, consistent with the theoretically predicted result. In this experiment, the relationship between period and angle was expressed by the following model Eq. [4].

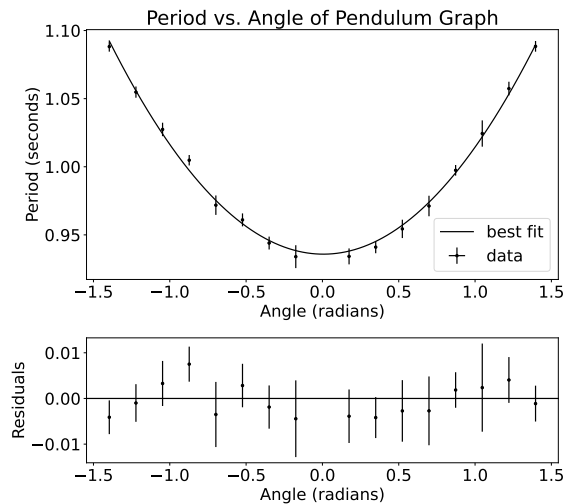


FIG. 3. Period vs. Angle graph based on 16 data points, including uncertainty error bars, quadratic best-fit curve, and residuals. Data are fit to Eq. [4].

The coefficients obtained through curve fitting were

$$T_0 = 0.936 \pm 0.002 \text{ s}$$

$$B = -0.001 \pm 0.001$$

$$C = 0.080 \pm 0.002$$

Theoretically, since the pendulum's motion is symmetric, B should be 0. In this experiment, B is equal to -0.001 ± 0.001 , which matches with zero within the uncertainty range. This indicates that the pendulum is mostly symmetric, and deviation is negligible. However, minor

uncertainties may arise from mechanical asymmetries.

Conversely, $C = 0.080 \pm 0.002$ is larger than zero, indicating a period dependence on the square of the angle. This reflects that the period increases as the release angle increases, which demonstrates that the small-angle approximation is not true at larger amplitudes.

To determine the range for the small-angle approximation, the following equation was applied.

$$T_0 C \theta_0^2 < U(T_0)$$

The $U(T_0)$ means the uncertainty of T_0 which is 0.002 s. After Substituting the values gives

$$\therefore -0.17 \text{ rad} < \theta_0 < +0.17 \text{ rad}$$

Within this range, the $C\theta_0^2$ remains smaller than the measurement uncertainty to confirm that the small-angle approximation is valid experimentally for angles below $\pm 0.17 \text{ rad}$ ($= 9.74^\circ$).

In summary, the results confirm that the small-angle approximation is valid within a small angular range of $\pm 0.17 \text{ rad}$ ($= 9.74^\circ$), but it breaks down at larger angles.

3.2. Q-Factor

The first method for determining the Q-factor involves observing the damping process of the pendulum and manually counting the number of oscillations until the amplitude decreases to a specific fraction of its initial value. To evaluate this, the Q/4 method derived from Eq. [3], the number of oscillations required for the amplitude to decay to approximately 46% ($e^{-\pi/4}$) of its initial value was measured. The initial release angle of 30° (0.52 rad) decreases to approximately 13.8° when reduced to 46% of its initial amplitude. From manual counting, the total number of oscillations was found to be $N = 148 \pm 2$. The ± 2 uncertainty was estimated based on possible counting errors during manual tracking. Therefore, substituting the measured value of N into Eq. [3], the Q-factor was obtained as shown below.

$$Q = 592 \pm 8$$

While this method is relatively simple and intuitive; however, it relies on lengthy video tracking and manual counting, which possibly resulting in high uncertainty.

The second method involved determining the decay constant τ by fitting the Time vs. Amplitude graph (FIG. 4) data to the damped harmonic oscillator model in Eq. [1]. Python-based nonlinear curve-fitting was used to obtain the data and perform the parameter estimation. The code was generated with the assistance of AI tools^[8], and the decay constant τ was extracted directly from the data.

In particular, the fitting process extracted the decay constant as

$$\tau = 178 \pm 1 \text{ s}$$

Using the measured average period of the pendulum from Period vs. Angle Section,

$$T_0 = 0.936 \pm 0.002 \text{ s}$$

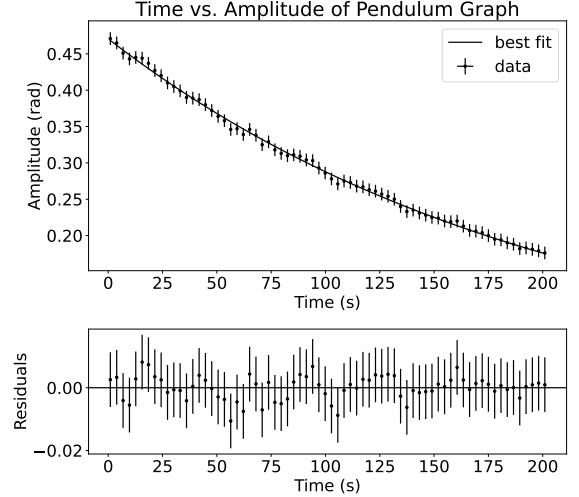


FIG. 4. Time vs. Amplitude graph of the pendulum released from 0.52 radian, fitted with an exponential curve. Residuals are shown below. Data are fit to the damped harmonic oscillator model in Eq. [1]

The quality factor was then calculated according to Eq. [2]:

$$Q = 597 \pm 5$$

This result matches well with the value obtained by the counting oscillation method ($Q = 592 \pm 8$), but the uncertainty is significantly smaller. Therefore, the curve fitting method provides a more reliable estimate of the Q-factor for the pendulum.

3.3. Period vs. Length Relation

Period vs. Length and Log (Period) vs. Log (Length) graphs (FIG. 5 and FIG. 6) illustrate how the oscillation period of a pendulum depends on its length.

To determine the parameters, the experimental dataset was fitted to the Power Law Model, Eq. [5] using Python-based nonlinear curve-fitting. This code was generated with the assistance of AI tools^[8], and the coefficients were extracted directly from the data. The resulting best-fit coefficients were:

$$k = 1.94 \pm 0.02$$

$$n = 0.433 \pm 0.004$$

To validate this relationship more closely, a log transformation was applied to both axes to obtain a linearized form, and the slope and intercept were calculated as follows:

$$n = 0.432 \pm 0.005$$

$$\log_{10}(k) = 0.288 \pm 0.003$$

$$k = 1.94 \pm 0.02$$

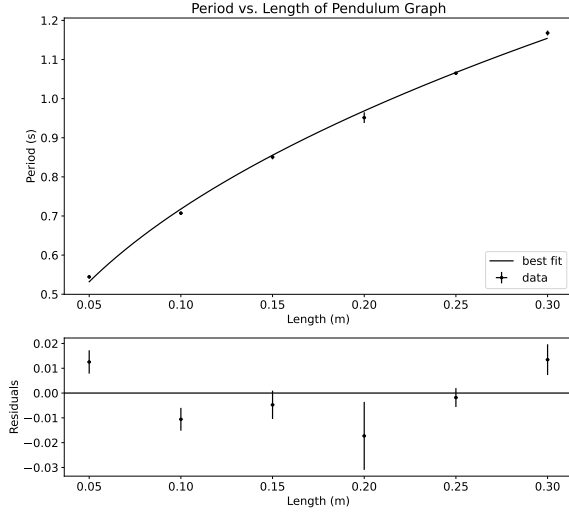


FIG. 5. Period vs. Length graph based on 6 data points, including uncertainty error bars, nonlinear best-fit curve, and residuals. Data are fit to the Power Law Model in Eq. [5].

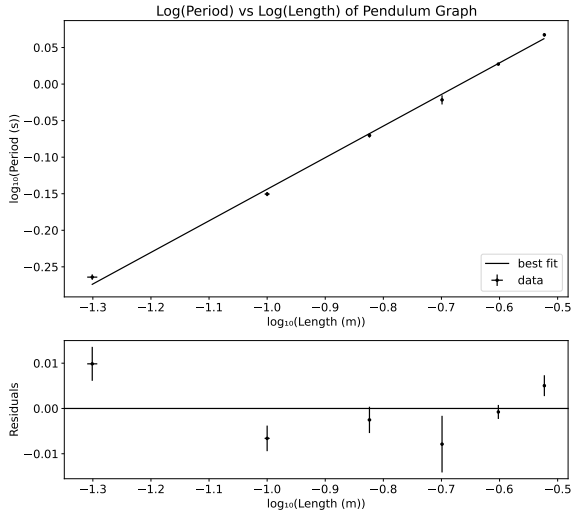


FIG. 6. Log (Period) vs. Log (Length) graph based on 6 data points, including uncertainty error bars, linear best-fit curve, and residuals. Linear fit to $\log_{10}(T) = \log_{10}(k) + n \log_{10}(L)$ (slope n , intercept $\log_{10} k$).

Theoretically, the Power Law Model Eq. [5], predicts $n = 0.5$ and $k = 2.0$ under the small-angle approximation. Comparing these two theoretical values and measurement values, k has 2.9% and n has 13.6% differences.

The causes of deviation in n included air resistance, friction, and uncertainties from frame analysis, which may occurred uncertainty factor. However, the most significant factor in this experiment was the violation of the small-angle approximation.

In the Period vs. Angle section, the small-angle approximation was calculated as

$$\therefore -0.17 \text{ rad} < \theta_0 < +0.17 \text{ rad}$$

However, in this experiment, the pendulum was released from 0.52 rad, which is far beyond the valid range of the small-angle approximation. At larger angles, the restoring force no longer follows the small-angle approximation, causing the period to increase more gradually with length than theoretically predicted. This explains why the experimental values deviate from the theoretical model.

3.4. Q-Factor vs. Length

The Q-Factor vs. Length graph (FIG. 7) presents how the damping behavior of the pendulum varies with length.

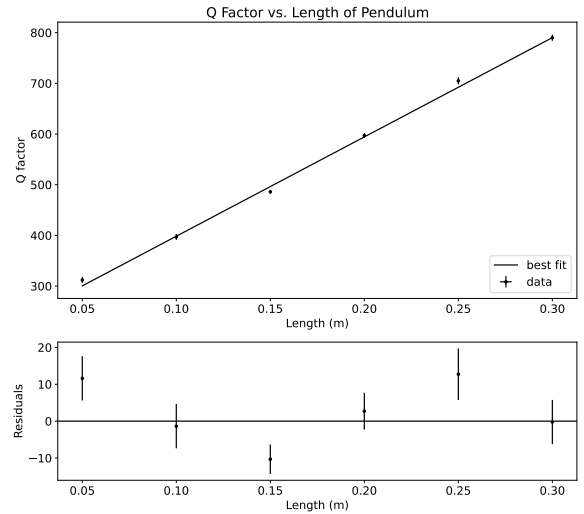


FIG. 7. Q-Factor vs. Length graph based on 6 data points, including uncertainty error bars, linear best-fit line, and residuals. The trend is fit to Eq. [7].

For each length, the amplitude decay was observed using the Tracker Tool with as following the same methods on the Time vs. Amplitude section. The τ was extracted by fitting the Time vs. Amplitude data to the damped harmonic oscillator model in Eq. [1]

The Q-factor for each length was then calculated using the model Eq. [2].

All Q-factors were obtained using curve fitting rather than manual counting to ensure minimized human error. The total of six lengths and Q values obtained, and considering the linear dependence between Q value and length, they were modeled as Eq. [7].

The fit analyzes results confirmed the following parameters.

$$a = 1960 \pm 30$$

$$b = 202 \pm 5$$

The results indicate that the Q-factor increases proportionally with length, which means longer pendulums experience relatively less damping.

This can also be interpreted through the theoretical relationship with model Eq. [6].

To verify this trend quantitatively, the theoretical period from Eq. [6] was inserted into Eq. [2], and the theoretical Q value was obtained as shown below.

$$Q = 594 \pm 16$$

The theoretical damping model Q value is consistent with the experimental value obtained from experiment 1 ($Q = 597 \pm 5$) within the range of uncertainty. This reflects the validity of the linear relationship between Q value and length, confirming that the damping in the pendulum decreases as length increases.

4. CONCLUSION

In this experiment, we examined the connection between the pendulum's damping characteristics using the Q-factor and its period and initial release angle. The period-angle graph confirmed that in the small-angle range, the period is independent of the initial amplitude, which is consistent with the small-angle approximation. However, in the large angle range, a large increase in the period was observed, clearly demonstrating the limitations of the small-angle approximation. From the fitted parameters in Eq. [4], the values $T_0 = 0.936 \pm 0.002$ s, $B = -0.001 \pm 0.001$, and $C = 0.080 \pm 0.002$ were obtained. The B value matched zero within the uncertainty range, confirming the pendulum motion was nearly symmetric. Conversely, the large positive value of C appears to show the expected dependence at large amplitudes.

For the damping analysis, two independent methods were used to determine the Q-factor. The manual counting method yielded $Q = 592 \pm 8$, while the curve fitting method applied to the Time vs. Amplitude data yielded $Q = 597 \pm 5$. Both results met in the uncertainty range to confirm internal consistency. Although the curve fitting method provided more precise estimates with decreased statistical uncertainty.

In the Period vs. Length and log-log analysis, the best-fit parameters for the Power Law Model, Eq. [5], give $k = 1.94 \pm 0.02$ and $n = 0.433 \pm 0.004$. Compared to the theoretical value of $k = 2.0$ and $n = 0.5$, the difference in the n value (approximately 13%) was due to the violation of the small-angle approximation. The small-angle approximation for this pendulum was ± 0.17 rad, but eventually the pendulum was released from 0.52 rad, which is far beyond the approximation. At large angles, the restoring force no longer follows the linear approximation, so the measured period grows slowly with increasing length than the theoretical $\sqrt{L}$ dependence, which lowers the fitted exponent n . The Q-factor vs. Length analysis further demonstrated that longer pendulums experience smaller effective damping,

consistent with the extracted linear relationship in Eq. [7]. Substituting the theoretical period yielded $Q = 594 \pm 16$, which overlaps with the results from experimental 1. This fact confirms that the data are consistent within uncertainty.

This experiment matched the theoretical predictions, but several limitations of the handmade apparatus affected precision. Placing a 1.5kg laptop on the acrylic base reduced the vibration, but residual vibration still could potentially shift the pivot during operation because the base was not rigidly fixed. Since the period depends on $\sqrt{L}$, even minor pivot movements can change the length, causing systematic variations in the measured period. Additionally, due to limitations of the recording setting, the video was recorded at 60 fps, resulting in a time resolution of $1/60$ s ≈ 16.7 ms. This resolution resulted in a non-negligible portion of period uncertainty, particularly for short vibrations. Furthermore, frame capacity constraints due to insufficient performance of the computer device limited the Tracker software to analyzing only the first 200 seconds of the approximately 13 minutes of decay, which further restricted the decay analysis and reduced the precision of the extracted decay constant τ . Finally, since the protractor was manually attached to the base with double-sided tape, it was not perfectly aligned between the geometrical origin and the actual pivot point, resulting in unavoidable minor angular setting errors.

Future experiments will yield more precise results by addressing limitations. Using a rigid mounting system can keep the rotation axis fixed, such as wall-mounted brackets or heavy metal blocks, to reduce the movement of the rotation axis. Furthermore, recording at higher frame rates (120–240 fps) can greatly reduce timing quantization errors. Furthermore, regarding issues encountered when using Tracker software, using software optimized for long-term tracking or by using a higher-performance computer device can fit the entire decay curve, enabling more precise calculation of the τ value. Finally, directly integrating a printed protractor into the pivot structure would improve the repeatability of the angular alignment and reduce the angular uncertainty. These improvements would address the limitations observed in this setup, resulting in more precise measurement results.

Overall, this experiment successfully examined the small-angle approximation, the damping behavior of harmonic oscillators, and the period-length dependence. Simultaneously, it clearly demonstrated the impact of the mechanical constraints arising from the handmade apparatus on the measured values, emphasizing the importance of precise experimental design and controlled measurement conditions.

REFERENCES

- [1][2][3][4][5][6] Wilson, Brian, "PHY180 Lab Project", downloaded from Quercus, 2024.
- [8] ChatGPT (OpenAI) provided assistance in creating Python code used for fitting models and extracting experimental coefficients.